Zihao Wang

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EDUCATION

Nanyang Technological University, Singapore **January 2025 - present**

Ph.D. candidate, Medicine (GPA: 4.88/5.0)

Advisors: Asst. Prof. Jue Tao Lim, Dr. Ian Wee Liang En (Singapore General Hospital)

Collaborators: Prof. David N. Fisman (University of Toronto), Assoc. Prof. Joseph Lewnard (University of California, Berkeley), Asst. Prof. Lilith K. Whittles (Imperial College London), Asst. Prof. Sen Pei (Columbia University)

- Research interests: Systems Modeling, Statistical Inference, Deep Learning.
- [Google Scholar].

National University of Singapore, Singapore **August 2019 - June 2023**

B.Eng. (Hons), Computer Engineering (Distinction, GPA: 4.35/5.0)

Advisors: Dr. Ganesh Neelakanta Iyer, Dr. Ravi Suppiah, Asst. Prof. Deepu John (University College Dublin)

- University Town College Program

PUBLICATIONS

1. Lin Geng*, Perran A. Ross*, Jo Yi Chow, **Zihao Wang**, Esther Li Wen Choo, Chia-Chen Chang, Xinyue Gu, Ary Hoffmann, Jue Tao Lim, "*Wolbachia*-mediated incompatible-insect technique robust to future climate change scenarios," submitted.
2. An Ting Tay*, Peihong Guo*, Jinghao Nicholas Ngaim*, Zihao Wang*, Liang En Wee, Calvin Chiew, Lalitha Kurupatham, Lee Ching Ng, Po Ying Chia, David Chien Boon Lye, Kelvin Bryan Tan, Jue Tao Lim*, "Risk of post-acute multi-organ sequelae after dengue reinfection," submitted.
3. Liang En Wee*, Reen Wan Li Ho*, **Zihao Wang***, Peihong Guo*, Rhea Khanna, Calvin Chiew, Russell Jingxian Li, Jiahui Li, Chee-Fu Yung, Chia Yin Chong, Zubaidah Said, David Chien Boon Lye, Vernon Lee, Kelvin Bryan Tan, Jue Tao Lim, "Acute and post-acute healthcare utilization following paediatric hospitalization for respiratory-syncytial-virus versus influenza or Omicron COVID-19 in a tropical setting: a population-based cohort study," submitted.
4. Peihong Guo*, Reen Wan Li Ho*, **Zihao Wang***, Ian Liang En Wee*, Russ Li, David Chien Boon Lye, Kelvin Bryan Tan, Jue Tao Lim, "Acute and post-acute healthcare utilization after hospitalization for COVID-19, influenza or respiratory syncytial virus in adults: a population-based cohort study," submitted.
5. **Zihao Wang**, Dariya Nikitin, Borame L. Dickens, Liang En Wee, Martin T.W. Chio, Rayner Kay Jin Tan, Keisuke Ejima, Yi Wang, David N. Fisman, Lilith K. Whittles, and Jue Tao Lim, "Long-term public health impact of doxycycline post-exposure prophylaxis on syphilis transmission," ***Nature Health***, 1-13, 2026. [Paper, GitHub].
6. **Zihao Wang**, Yuhan Li, Yao Shi, and Zhe Wu, "Input convex lipschitz recurrent neural networks for robust and efficient process modeling and optimization," arXiv. [arXiv, GitHub].
7. **Zihao Wang**, and Zhe Wu, "Towards foundation model for chemical reactor modeling: Meta-learning with physics-informed adaptation," ***Chemical Engineering Research and Design*** 218, 839-853, 2025. IF: 3.9, JCR Q2 (Engineering, Chemical). [Paper, GitHub].
8. **Zihao Wang**, Donghan Yu, and Zhe Wu, "Real-time machine-learning-based optimization using input convex long short-term memory network," ***Applied Energy*** 377, 124472, 2025. IF: 11.0, JCR Q1 (Engineering, Chemical; Energy & Fuels). [Paper, GitHub].
9. **Zihao Wang**, and Ravi Suppiah, "Upper limb movement recognition utilising EEG and EMG signals for rehabilitative robotics," *Proceedings of Future of Information and Communication Conference*, 676-695, San Francisco, California, USA, 2023. [Paper, GitHub].

   (* denotes equal contribution)

PRESENTATIONS

1. **Zihao Wang***, Dariya Nikitin, Borame L. Dickens, Liang En Wee, Martin T.W. Chio, Rayner Kay Jin Tan, Keisuke Ejima, Yi Wang, David N. Fisman, Lilith K. Whittles, and Jue Tao Lim, "Long-term public health impact

of doxycycline post-exposure prophylaxis on syphilis transmission," *Singapore HIV, Hepatitis and Sexually Transmitted Infections Congress (SHHSC)*, Singapore, 2026. (**Oral)**.
2. **Zihao Wang***, Dariya Nikitin, Liang En Wee, Martin T.W. Chio, Rayner Kay Jin Tan, Yi Wang, David N. Fisman, Lilith K. Whittles, and Jue Tao Lim, "Modelling the interplay between doxycycline post-exposure prophylaxis, vaccination, and antimicrobial resistance in *Neisseria gonorrhoeae*," *European Congress of Clinical Microbiology and Infectious Diseases (ESCMID Global 2026)*, Munich, Germany, 2026. (Poster, **Top 2%**).
3. **Zihao Wang***, Dariya Nikitin, Liang En Wee, Martin T.W. Chio, Rayner Kay Jin Tan, Yi Wang, David N. Fisman, Lilith K. Whittles, and Jue Tao Lim, "Modelling the interplay between doxycycline post-exposure prophylaxis, vaccination, and antimicrobial resistance in *Neisseria gonorrhoeae*," *LKCMedicine EArly Researcher Network (LEARN) Symposium*, Singapore, 2026. (**Flash Talk**).
4. **Zihao Wang***, Dariya Nikitin, Borame L. Dickens, Liang En Wee, Martin T.W. Chio, Rayner Kay Jin Tan, Keisuke Ejima, Yi Wang, David N. Fisman, Lilith K. Whittles, and Jue Tao Lim, "Long-run public health impact of doxycycline post-exposure prophylaxis and behavioural factors on syphilis transmission: A modelling study in Singapore and England," *Epidemics: 10th International Conference on Infectious Disease Dynamics (Epidemics10)*, San Diego, California, USA, 2025. (**Oral)**.
5. **Zihao Wang***, Dariya Nikitin, Borame L. Dickens, Liang En Wee, Martin T.W. Chio, Rayner Kay Jin Tan, Keisuke Ejima, Yi Wang, David N. Fisman, Lilith K. Whittles, and Jue Tao Lim, "Long-term public health impact of doxycycline post-exposure prophylaxis and behavioural factors on syphilis transmission: A modelling study in Singapore and England," *Singapore Health & Biomedical Congress (SHBC)*, Singapore, 2025. (Poster).
6. **Zihao Wang**, Wenlong Wang*, and Zhe Wu, "Accelerated modeling of various chemical processes using meta-learning-based foundation models: A few-shot learning approach using Reptile," *American Institute of Chemical Engineers (AIChE) Annual Meeting*, San Diego, California, USA, 2024. (**Oral)**. [Abstract].
7. **Zihao Wang**, Wenlong Wang*, and Zhe Wu, "Input convex LSTM for fast machine learning-based optimization," *American Institute of Chemical Engineers (AIChE) Annual Meeting*, San Diego, California, USA, 2024. (Poster). [Abstract].
8. **Zihao Wang**, and Zhe Wu*, "Input convex LSTM: A convex approach for fast model predictive control," Institute for Machine Learning (Head: Professor Sepp Hochreiter), Johannes Kepler University, Linz, Austria, December 2023. (**Invited talk** by Professor Günter Klambauer).

(* denotes presenting author)

AWARDS AND HONORS

Travel Grant, ESCMID Global 2026 — **04/2026**
Top Rated Abstract, ESCMID Global 2026 — **04/2026**
NTU Research Scholarship, NTU — **2025 - Present**
IEEE Singapore Computer Society Prize, NUS — **06/2023**
Outstanding Computing Project Prize, NUS — **06/2023**

DISSERTATIONS

1. **Zihao Wang**, "Multimodal sensor fusion using AI for healthcare IoT sensors," B.Eng. Dissertation, Computer Engineering, National University of Singapore, Singapore, 2023. [Paper].
2. **Zihao Wang**, "AI research for rehabilitative robotics," Undergraduate Research Opportunity Project (UROP) Thesis, Computer Engineering, National University of Singapore, Singapore, 2022. [Paper].

PROFESSIONAL EXPERIENCES

National Centre for Infectious Diseases — **May 2025 - May 2026**
Research Staff (Part-time), Tan Tock Seng Hospital, Singapore
Advisors: Assoc. Prof. Kelvin Bryan Tan, Asst. Prof. Jue Tao Lim

- Conduct research on evaluating different methodological approaches to estimate the true infection fatality ratio (IFR) during pandemics, with the aim of informing early estimation of IFR under limited data to support timely, evidence-based policy decisions, under the Programme for Research in Epidemic Preparedness and REsponse (PREPARE)

- Conduct research on population-based retrospective cohort studies for respiratory syncytial virus (RSV) and dengue

Department of Chemical & Biomolecular Engineering **July 2023 - January 2025**
Research Engineer (Full-time), National University of Singapore, Singapore
Advisor: Asst. Prof. Zhe Wu

- Conducted research on input convex neural networks, model predictive control, chemical process modeling, meta-learning, and physics-informed learning
- Wrote concept papers and full proposals for grant applications and supervised master's students

Department of Biomedical Engineering **March 2023 - July 2023**
Undergraduate Research Assistant (Internship), National University of Singapore, Singapore
Advisor: Asst. Prof. Yueming Jin

- Conducted research on computer vision for gastrectomy surgical video analysis and modality-cataract video segmentation

Rolls-Royce@NTU Corporate Laboratory **May 2022 - August 2022**
Research Engineer (Internship), Nanyang Technological University, Singapore
Advisors: Dr. Aditya Venkataraman and Dr. Benjamin Cheong

- Worked on software engineering solutions for hybrid-electric power and propulsion design tools for marine and aero engines

NUS Human-Computer Interaction (NUS-HCI) Lab **March 2022 - May 2022**
Undergraduate Research Assistant (Internship), National University of Singapore, Singapore
Advisor: Prof. Shengdong Zhao

- Conducted empirical research on human-computer interaction

SKILLS

Computer: C, C++, Java, Python, R, Assembly Language, MATLAB, PostgreSQL, Verilog, Vensim, Gephi, LaTeX
Languages: Mandarin (native), English (bilingual), Cantonese (proficient), Korean (conversational)

Appendix:

PUBLICATIONS

1.

PROFESSIONAL EXPERIENCES

National University of Singapore Singapore
Research Engineer (Full-time) July 2023 - January 2025

- Proposed a novel **Input Convex LSTM** with an average speedup for neural network-based optimization and control of 2x on a nonlinear chemical reactor and 4x on a solar PV hybrid energy system at LHT Holdings in Singapore, implemented using Tensorflow Keras and IPOPT
- Proposed a novel **Input Convex Lipschitz RNN** for fast and robust neural network-based model predictive control on nonlinear chemical reactors and Organic Rankine Cycle-based waste heat recovery systems, implemented using Tensorflow Keras and IPOPT
- Researched in meta-learning (i.e., Reptile) and physics-informed learning to develop a **Foundation Model** for chemical reactor modelling, which enables few-shot adaptation to any unseen chemical processes in the reactors (including unseen reactor types), implemented using Tensorflow Keras
- Wrote concept paper and full proposal for grant application (**NUS Foresight Grant**) with grant amount of **S$ 1 million**, centred around my research topic on foundation model for chemical reactor modelling

Rolls-Royce@NTU Corporate Laboratory Singapore
Research Engineer (Internship) May 2022 - August 2022

- Developed and maintained hybrid-electric power and propulsion design tools for marine and aero engines
- Programmed a multi-windows app using MATLAB App Designer for the evaluation of different hybrid-electric marine engine designs, consisting of Gurobi for mathematical optimisation
- Configured layered architecture for a Python-based design environment for the evaluation of different hybrid-electric aircraft architectures, and built surrogate models using Bivariate Spline interpolator and Gaussian Process regressor for aerodynamic analysis

SELECTED PROJECTS

NUS Department of Biomedical Engineering Singapore
Undergraduate Research Assistant (Internship) March 2023 - July 2023

- Annotated Gastrectomy surgical video and Modality-cataract video to provide region masks for the surgeons using Labelbox and Computer Vision Annotation Tool (CVAT) respectively
- Conference Assistant at the 6th IEEE-RAS International Conference on Soft Robotics (*RoboSoft*)

Multimodal Data Fusion for Sleep Apnea Detection, Final Year Project [GitHub] Singapore
Individual August 2022 – May 2023

- Proposed a novel feature-level dynamic multimodal data fusion framework (DynNet) aimed at reducing computational complexity, offering a notable improvement over existing static multimodal data fusion techniques
- Evaluated the robustness and efficacy of DynNet in sleep apnea classification using real-time Electrocardiography (ECG) and Photoplethysmography (PPG) signals in the presence of varying signal-to-noise ratio of Gaussian noise, implemented using TensorFlow Keras

NUS Human-Computer Interaction (NUS-HCI) Lab Singapore
Undergraduate Research Assistant (Internship) March 2022 - May 2022

- Investigated the importance of Self-Initiated Explorable Instructions and the effect of text spacings in on-the-go reading using Optical See-Through Head-Mounted Displays through empirical research

Human Activity Recognition, Computer Engineering Capstone Project [GitHub] Singapore

Team Leader August 2021 – November 2021

- Constructed a real-time system from scratch by analysing EMG signals, accelerometer, and gyroscope readings for human activity recognition, where the system is built to classify different dance moves performed by multiple dancers simultaneously and detect dancers' real-time relative positions
- Overlooked the project, collected data, designed an algorithm for relative position detection, trained a multilayer perceptron model for multi-class classification and implemented it on FPGA, fused all subparts to form a holistic system, and achieved a final accuracy of 96.9% among 9 real-time activities

Upper Limb Movement Recognition, Undergraduate Research Opportunities Program [GitHub] Singapore
Individual August 2021 – May 2022

- Researched in upper limb movement multi-class classification using real-time Electroencephalography (EEG) and Electromyography (EMG) signals with decision-level data fusion for rehabilitative robotics control, implemented using PyTorch

Real Time Operating Robot [GitHub] Singapore
Team Leader August 2020 – November 2020

- Constructed a robot with FRDM KL25Z Board (ARM), BT06 module, buzzer, and LEDs such that the operator can establish a Bluetooth connection and tele-operate it via an Android application for the robot, buzzer, and LEDs to work concurrently
- Assembled the robot, designed a multithreaded algorithm for robot movement, buzzer, and LEDs display

Alex to the Rescue [GitHub] Singapore
Team Leader January 2020 – April 2020

- Constructed a robot with Raspberry Pi, Arduino UNO, and LiDAR such that the operator can tele-operate it via command window to search and rescue the target in an unknown maze and visualise the structure of the maze simultaneously
- Assembled the robot, designed an algorithm for robot movement, configured communications among Raspberry Pi, Arduino UNO, and LiDAR

The A-maze-ing Race Project [GitHub] Singapore
Team Leader August 2019 – November 2019

- Constructed a robot with Arduino UNO, sound sensor, ultrasonic sensor, infra-red sensor, and colour sensor such that it would be able to pass through an unknown maze autonomously
- Assembled the robot, designed an algorithm for robot movement, implemented infra-red sensor and ultrasonic sensor for distance measurement, implemented sound sensor for instruction intake